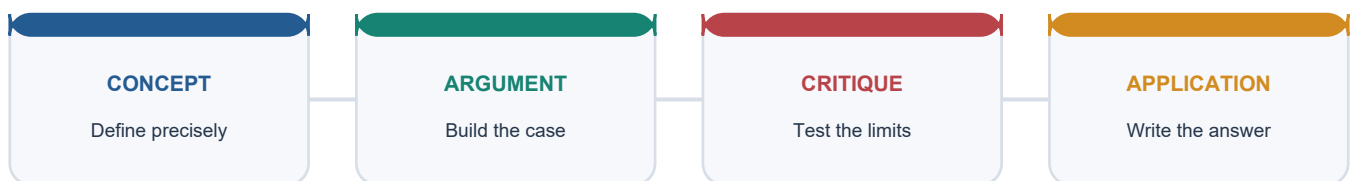


COMPLETE LEARNING SESSION

Economy Topic 27 - Digital Agriculture, Agritech and e-Technology for Farmers - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Economy Topic 27 - Digital Agriculture, Agritech and e-Technology for Farmers - Solved Practice Workbook

Current-source cutoff: 10 September 2026.

All 32 MCQs precede PYQs. Correct answers rotate A-B-C-D exactly eight times. Every option has a question-specific explanation and each item records a unique trap.

32 SOLVED MCQS

MCQ 01

Question: Which statement best distinguishes digital agriculture from agritech?

- A. Digital agriculture is data-enabled farm-cycle management, while agritech is the wider technology and enterprise ecosystem
- B. Agritech means only government portals
- C. Digital agriculture means mechanisation without data
- D. Both terms mean precision irrigation alone

Correct answer: A

- **Option A:** The first option preserves the narrower data-management concept and wider ecosystem boundary.
- **Option B:** Private firms, devices and business models also belong to agritech.
- **Option C:** Mechanisation may be non-digital and digital agriculture extends beyond machines.
- **Option D:** Precision irrigation is one application, not the entire field.

Unique trap: Do not reduce the topic to one portal or device.

MCQ 02

Question: Which sequence correctly represents the welfare transmission chain?

- A. App download to registration to welfare
- B. Reliable data to validated advice to feasible action to measured outcome
- C. Satellite image directly to higher income
- D. Farmer ID directly to land title

Correct answer: B

- **Option A:** Activity counts omit decision quality and implementation.
- **Option B:** The complete chain includes validation, action and outcome feedback.
- **Option C:** Imagery supplies evidence, not an automatic causal result.
- **Option D:** A service identifier does not confer property rights.

Unique trap: Data availability is not farmer welfare.

MCQ 03

Question: Precision agriculture is best understood as:

- A. Any agricultural DBT
- B. Any use of a smartphone
- C. Spatially or temporally variable treatment based on measured need
- D. Universal ownership of advanced machinery

Correct answer: C

- **Option A:** A transfer can be digital but is not precision management.

- **Option B:** A device does not establish variable treatment.
- **Option C:** This is the defining management logic.
- **Option D:** Shared services may be more economical than ownership.

Unique trap: Precision is a treatment logic, not generic digitisation.

MCQ 04

Question: Which limitation most directly applies to remote sensing?

- A. It can never observe crop condition
- B. It is identical to field inspection
- C. It provides land title
- D. Cloud, resolution and crop similarity require ground truth

Correct answer: D

- **Option A:** Remote sensing is widely used for crop observation.
- **Option B:** Field inspection and remote inference are complementary.
- **Option C:** Imagery does not determine legal title.
- **Option D:** These physical and classification limits justify validation.

Unique trap: A scalable signal is not automatic ground truth.

MCQ 05

Question: Which pairing is correct?

- A. GIS analyses spatial layers; GNSS/NavIC supplies positioning signals
- B. GIS is an Indian navigation constellation
- C. NavIC is a crop-classification algorithm
- D. GPS is a cadastral land record

Correct answer: A

- **Option A:** The pair separates analysis from positioning.
- **Option B:** NavIC, not GIS, is the Indian regional navigation system.
- **Option C:** Navigation signals do not themselves classify crops.
- **Option D:** GPS is a positioning system, not a legal record.

Unique trap: GIS, GNSS and remote sensing have distinct functions.

MCQ 06

Question: An agricultural drone's task depends primarily on:

- A. Airframe colour
- B. Payload, configuration, software and trained operation
- C. Farmer ID count
- D. The e-NAM commodity list

Correct answer: B

- **Option A:** Colour does not determine capability.
- **Option B:** Mapping, sensing or spraying requires the relevant complete system.
- **Option C:** A registry count is unrelated to aircraft function.
- **Option D:** Market listings do not configure drones.

Unique trap: The platform alone does not supply every capability.

MCQ 07

Question: Which statement correctly separates IoT and AI?

- A. AI is a soil sensor
- B. IoT always makes autonomous decisions
- C. IoT captures/exchanges data while AI infers or predicts from data
- D. They are legally identical

Correct answer: C

- **Option A:** A sensor is a data-capture device.
- **Option B:** Connected devices need not automate action.
- **Option C:** This preserves sensing and inference layers.
- **Option D:** The technologies and accountability roles differ.

Unique trap: Sensing, inference and control are separate layers.

MCQ 08

Question: A high aggregate AI accuracy rate is insufficient because:

- A. Agriculture has no data
- B. All models are prohibited
- C. Only hardware matters
- D. Subgroup bias, drift and rare-event failure may remain

Correct answer: D

- **Option A:** Agriculture produces abundant but uneven data.
- **Option B:** Models are not generally prohibited; their use needs safeguards.
- **Option C:** Data and decision design matter alongside hardware.
- **Option D:** Distributional and temporal performance can differ from the average.

Unique trap: Accuracy does not equal fairness or robustness.

MCQ 09

Question: Which official fact is correctly stated?

- A. Cabinet approved the Digital Agriculture Mission on 3 September 2024 with Rs 2,817 crore outlay
- B. The Mission was completed nationwide in 2021
- C. Its outlay equals certified expenditure
- D. It is one mobile application

Correct answer: A

- **Option A:** The dated official release records the approval and outlay.
- **Option B:** IDEA was a 2021 blueprint, not nationwide completion.
- **Option C:** Sanctioned outlay is not realised spending.
- **Option D:** The Mission is an umbrella architecture.

Unique trap: Approved, spent and achieved are different stages.

MCQ 10

Question: IDEA 2021 is best described as:

- A. A final crop survey count
- B. A digital-ecosystem design framework preceding the 2024 Mission
- C. An insurance claim formula
- D. An e-NAM settlement rule

Correct answer: B

- **Option A:** Coverage is a later implementation metric.
- **Option B:** The chronology places blueprint before funded mission.
- **Option C:** PMFBY rules govern insurance claims.
- **Option D:** Market settlement has a separate architecture.

Unique trap: A blueprint is not an operational outcome.

MCQ 11

Question: AgriStack's official architecture is:

- A. A private input marketplace
- B. One central file controlled by e-NAM
- C. A federated Centre-state DPI built around farmer, map and crop registries
- D. A satellite constellation

Correct answer: C

- **Option A:** It is public digital infrastructure, not an input seller.
- **Option B:** e-NAM and AgriStack have distinct functions.
- **Option C:** This captures the official federated design.
- **Option D:** It uses geospatial data but is not a space system.

Unique trap: Federated does not automatically mean accurate or inclusive.

MCQ 12

Question: Farmer ID should not be treated as:

- A. A reusable service identifier
- B. A state-maintained registry key
- C. A route to authorised verification
- D. Conclusive land title and guaranteed entitlement

Correct answer: D

- **Option A:** Reusable verification is an intended function.
- **Option B:** States and UTs maintain the registry.
- **Option C:** Services may request authorised verification.
- **Option D:** Identity, title, cultivation and eligibility are distinct.

Unique trap: Identity is not substantive eligibility.

MCQ 13

Question: The Digital Crop Survey primarily records:

- A. Plot-level crop-sown information for a season
- B. Only bank transactions
- C. Only satellite images without field capture
- D. Legal title transfers

Correct answer: A

- **Option A:** The mobile field workflow links parcel, crop and season.
- **Option B:** Financial transactions are a different data class.
- **Option C:** Remote sensing may complement field survey.
- **Option D:** Revenue authorities and legal processes govern title.

Unique trap: Surveyed plots are not automatically benefited farmers.

MCQ 14

Question: Krishi-DSS is best classified as:

- A. An autonomous authority
- B. A geospatial decision-support platform combining multiple datasets
- C. A drone licence
- D. A direct-income transfer

Correct answer: B

- **Option A:** Accountable institutions retain decision authority.
- **Option B:** The platform integrates satellite, weather, soil, crop and water layers.
- **Option C:** DGCA and aviation rules govern drone permissions.
- **Option D:** PM-KISAN is a separate DBT scheme.

Unique trap: Decision support is not automated legal authority.

MCQ 15

Question: Open standards in agricultural DPI mean:

- A. All personal data must be public
- B. One vendor owns every interface
- C. Systems can interoperate under governed access rules
- D. Consent and security are unnecessary

Correct answer: C

- **Option A:** Openness of standards differs from public disclosure.
- **Option B:** Open interfaces should reduce vendor dependence.
- **Option C:** Technical compatibility operates with legal controls.
- **Option D:** Privacy and security remain mandatory design questions.

Unique trap: Interoperability is not unrestricted access.

MCQ 16

Question: What is the correct DPDP status by 10 September 2026?

- A. Only a draft Act exists
- B. All provisions commenced in August 2023
- C. Agriculture is entirely exempt
- D. Rules were notified 13 November 2025 with phased commencement

Correct answer: D

- **Option A:** The Act was enacted in 2023.
- **Option B:** Enactment and operative commencement differ.
- **Option C:** No blanket agriculture exemption follows.
- **Option D:** The Gazette notifications use immediate, one-year and eighteen-month groups.

Unique trap: Never merge enactment, notification and commencement.

MCQ 17

Question: Aadhaar authentication establishes:

- A. Identity credential verification, not title, cultivation or eligibility
- B. Ownership of every linked parcel
- C. Correct crop classification
- D. Automatic PM-KISAN payment

Correct answer: A

- **Option A:** The authentication perimeter is identity.
- **Option B:** Land rights require revenue and legal evidence.
- **Option C:** Crop survey and remote sensing establish crop information.
- **Option D:** Scheme conditions and bank validation still apply.

Unique trap: Authentication is only one verification layer.

MCQ 18

Question: The PM-KISAN portal's physical-verification warning shows that:

- A. Digital records are never used
- B. Digital checks require corrective field verification for suspected cases
- C. Every beneficiary is a tenant
- D. DBT itself determines eligibility

Correct answer: B

- **Option A:** The scheme uses extensive digital verification.
- **Option B:** The portal explicitly withholds suspected cases pending verification.
- **Option C:** Tenancy status varies and is not implied.
- **Option D:** Entitlement rules precede payment execution.

Unique trap: Digital delivery needs an exception-and-correction path.

MCQ 19

Question: Digital KCC processing should be understood as:

- A. Automatic grant finance
- B. A substitute for all bank underwriting
- C. A bank-credit workflow made more efficient by verified data
- D. A crop-insurance claim

Correct answer: C

- **Option A:** KCC is repayable credit.
- **Option B:** Banks retain appraisal and borrower-protection duties.
- **Option C:** This preserves digital efficiency and credit discipline.
- **Option D:** Insurance is a separate risk-transfer product.

Unique trap: Farmer ID does not guarantee sanction.

MCQ 20

Question: The central risk of alternative-data scoring is:

- A. It cannot use any data
- B. It eliminates information asymmetry completely
- C. It always lowers interest rates
- D. Proxy bias may cause exclusion or excessive lending

Correct answer: D

- **Option A:** Alternative data uses non-traditional signals.
- **Option B:** Uncertainty and asymmetry remain.
- **Option C:** Pricing depends on risk, competition and costs.
- **Option D:** Biased proxies and aggressive offers can harm borrowers.

Unique trap: Richer data can create new discrimination.

MCQ 21

Question: YES-TECH and WINDS belong to:

- A. Technology-enabled PMFBY yield/weather architecture
- B. e-NAM trading rules
- C. DGCA pilot licensing
- D. PM-KISAN land mutation

Correct answer: A

- **Option A:** The PMFBY framework uses technology for yield and weather evidence.
- **Option B:** e-NAM concerns agricultural marketing.
- **Option C:** DGCA governs aviation safety.
- **Option D:** Revenue records and PM-KISAN verification are separate.

Unique trap: Insurance technology does not eliminate basis risk.

MCQ 22

Question: Basis risk means:

- A. The insurer has no database
- B. Measured/index loss differs from the farmer's actual loss
- C. Every claim is fraudulent
- D. The premium is a land title

Correct answer: B

- **Option A:** Database quality is one operational issue.
- **Option B:** This is the defining mismatch.
- **Option C:** Disagreement may arise without fraud.
- **Option D:** Premium and title are unrelated concepts.

Unique trap: A faster model can still mismatch individual loss.

MCQ 23

Question: Which statement about e-NAM is correct?

- A. It abolishes state APMC law
- B. It physically transports every lot
- C. It is an SFAC-operated electronic market layer needing assaying, logistics and settlement
- D. It guarantees a higher price

Correct answer: C

- **Option A:** State market frameworks continue.
- **Option B:** Physical logistics remain separate.
- **Option C:** This captures the digital-plus-physical model.
- **Option D:** Price outcomes depend on competition, quality and execution.

Unique trap: Online bidding alone is not a completed trade.

MCQ 24

Question: FPO digitalisation can raise bargaining power only when:

- A. The app replaces members
- B. It avoids books and governance
- C. Every member owns a drone
- D. Aggregation, working capital, transparent governance and buyers function together

Correct answer: D

- **Option A:** Member agency remains essential.
- **Option B:** Records and accountability support collective action.
- **Option C:** Shared service may be preferable.
- **Option D:** Digital tools need institutional and market complements.

Unique trap: FPO registration is not active collective trade.

MCQ 25

Question: Why can drone-as-a-service outperform individual ownership?

- A. It spreads fixed cost across clustered demand
- B. It removes all regulation
- C. It makes maintenance unnecessary
- D. It guarantees year-round use

Correct answer: A

- **Option A:** Higher utilisation lowers cost per operation.
- **Option B:** Aviation and application rules still apply.
- **Option C:** Maintenance becomes a provider responsibility, not unnecessary.
- **Option D:** Agricultural demand remains seasonal.

Unique trap: Asset distribution and service utilisation differ.

MCQ 26

Question: Platform network effects may create which trade-off?

- A. No users benefit
- B. Better matching alongside concentration and lock-in risk
- C. All intermediaries disappear
- D. Competition law becomes irrelevant

Correct answer: B

- **Option A:** Users may gain from broader networks.
- **Option B:** The same scale can improve service and gatekeeping power.
- **Option C:** New digital intermediaries can emerge.
- **Option D:** Competition remains relevant.

Unique trap: Digital disintermediation can create a new intermediary.

MCQ 27

Question: Which cyber design is most farmer-protective?

- A. Online-only service with no recovery
- B. Shared passwords for convenience
- C. Strong authentication, logs, backups, incident response and offline continuity
- D. Permanent storage of every field

Correct answer: C

- **Option A:** Outages would cause exclusion.
- **Option B:** Shared credentials undermine accountability.
- **Option C:** Layered resilience addresses prevention and recovery.
- **Option D:** Data minimisation and retention limits remain important.

Unique trap: Backend security must be joined to usable recovery.

MCQ 28

Question: The strongest evaluation metric is:

- A. Number of app downloads
- B. Number of messages sent
- C. Number of IDs created
- D. Verified change in net income, risk or resource use with distribution and cost

Correct answer: D

- **Option A:** Downloads do not prove use.
- **Option B:** Delivery does not prove action.
- **Option C:** IDs are infrastructure outputs.
- **Option D:** Outcome, cost and distribution test welfare.

Unique trap: Activity metrics are not impact metrics.

MCQ 29

Question: Why can precision irrigation fail to reduce total groundwater extraction?

- A. Lower unit cost may induce area or crop expansion
- B. Sensors legally require more pumping
- C. GIS creates water
- D. Every farmer abandons irrigation

Correct answer: A

- **Option A:** This is the rebound mechanism.
- **Option B:** No such general legal requirement exists.
- **Option C:** Spatial analysis does not create resources.
- **Option D:** Behaviour varies.

Unique trap: Per-hectare efficiency is not basin-level conservation.

MCQ 30

Question: Why is state capacity central to AgriStack?

- A. The Union alone maintains all land records
- B. States maintain key registries and field surveys within common standards
- C. e-NAM updates every cadastral map
- D. Private apps confer title

Correct answer: B

- **Option A:** Land and implementation roles are federal and state-linked.
- **Option B:** This reflects the official federated architecture.
- **Option C:** e-NAM is a market platform.
- **Option D:** Private applications cannot create legal title.

Unique trap: National launch is not uniform state implementation.

MCQ 31

Question: Agricultural drone use requires:

- A. Only a smartphone
- B. Only an e-NAM account
- C. Aviation compliance plus trained operation and task/input-specific safety rules
- D. Only Farmer ID

Correct answer: C

- **Option A:** A phone is insufficient.
- **Option B:** Market registration is unrelated.
- **Option C:** The regulatory perimeters are cumulative.
- **Option D:** Identity does not authorise flight or spraying.

Unique trap: One certification does not replace another perimeter.

MCQ 32

Question: The best final test of digital agriculture is whether it:

- A. Maximises databases
- B. Eliminates human extension
- C. Centralises every service
- D. Expands farmer capability, income and resilience with control and remedy

Correct answer: D

- **Option A:** Database growth is an input.
- **Option B:** Human interpretation remains important.
- **Option C:** Federation and competition may be preferable.
- **Option D:** This balances outcomes and agency.

Unique trap: Digitisation is a means, not the objective.

PYQS AND OFFICIAL-KEY DISCIPLINE

VERIFIED MAINS PYQ - 2023 GS-III - 10 MARKS

Question: How does e-Technology help farmers in production and marketing of agricultural produce? Explain it. Answer in 150 words.

Model solution: e-Technology converts farm, weather and market data into decisions and services. In production, remote sensing and GIS support acreage, crop-health and drought assessment; IMD-linked advisories improve sowing and harvest timing; sensors, precision irrigation and drones can target water or crop protection; and digital credit, insurance and crop surveys reduce verification cost. In marketing, price information, e-NAM bidding, digital assaying, FPO aggregation, warehouse records, logistics matching and electronic settlement can widen discovery and reduce transaction cost. These benefits require physical complements: connectivity, local-language extension, suitable inputs, assaying, storage, transport, working capital and dispute resolution. Land-linked records may exclude tenants or women cultivators, while model error and platform power require correction, appeal, interoperability and competition. Therefore, e-Technology aids farmers when reliable data becomes an affordable action and improves net income or risk, not merely when an app is launched.

VERIFIED PRELIMS PYQ - 2020 GS-I Q42 - KEY NEUTRAL

Question: Consider the following activities: (1) Spraying pesticides on a crop field; (2) Inspecting the craters of active volcanoes; (3) Collecting breath samples from spouting whales for DNA analysis. At the present level of technology, which of the above activities can be successfully carried out by using drones?

Answer withheld pending official UPSC key.

Concept solution: All three are technically plausible drone applications with appropriate payloads and operating conditions. The package does not infer an official option letter because the matched final official key is not held in the verified local set.

ROUTED 2024 OBJECTIVE DEMAND

The 2024 GS-I paper tested the World Economic Forum's **100 Million Farmers** platform. The routed concept is a multistakeholder platform concerned with transition towards net-zero, nature-positive food and water systems and farmer resilience. It is not an Indian government AgriStack or e-NAM programme. Exact options and an official answer letter are not reproduced here without the matched paper-key mapping.

ORIGINAL MAINS PRACTICE

MAINS 1 - 10 MARKS

Question: Explain the data-to-decision-to-action chain through which digital agriculture can improve farmer welfare. Answer in 150 words.

Native-body ceiling: 150 words under the required alphanumeric regex.

Model answer:

Digital agriculture begins with reliable data on farmer, parcel, crop, soil, weather, prices and risk. Satellites, surveys, sensors or administrative records capture signals; validation and interoperable systems convert them into usable information. Decision tools and human extension then translate information into local advice. Welfare rises only if the farmer can act through affordable inputs, credit, machinery, insurance, logistics and a competitive market. Outcomes must be measured through net income, loss reduction, resource use, resilience and farmer control, not messages or registrations. Remote sensing still needs ground truth; AI requires local validation; and a registry error needs assisted correction and appeal. Feedback from realised yield, loss and price must improve the next recommendation. Thus, the decisive link is not data abundance but an accurate, affordable and contestable action supported by physical institutions.

MAINS 2 - 10 MARKS

Question: Why must AgriStack be governed as a federated and contestable public infrastructure? Answer in 150 words.

Native-body ceiling: 150 words under the required alphanumeric regex.

Model answer:

AgriStack links state-maintained Farmer Registry, geo-referenced village maps and crop-sown information through common standards. Federation fits India's allocation of land and agricultural administration and avoids describing the system as one central database. Common rails can reduce repeated verification and support credit, insurance, procurement or advisory. Yet federation alone does not ensure legitimacy. Land records may omit tenants, sharecroppers and women cultivators; seasonal crop entries may be wrong; and linked services can spread error quickly. Governance therefore requires clear purpose, minimum data, controlled access, security logs, portability, local-language assistance, time-bound correction and human appeal before denial. Independent audits should publish error, correction and exclusion patterns across states. Farmer ID must remain a service identifier, not title or guaranteed entitlement. AgriStack succeeds when reusable verification expands farmer capability without making an inaccurate record an automated gatekeeper.

MAINS 3 - 15 MARKS

Question: Assess the potential and limitations of precision agriculture for Indian smallholders. Answer in 250 words.

Native-body ceiling: 250 words under the required alphanumeric regex.

Model answer:

Precision agriculture varies water, seed, nutrients or crop protection by measured spatial or temporal need. Soil-moisture sensors, GPS-guided equipment, remote imagery, drones and decision tools can improve timing, reduce avoidable input use, identify stress early and raise quality. The gains can lower cost and risk, especially when combined with weather advice and extension. Indian smallholders, however, face fragmented plots, seasonal demand, weak connectivity, limited capital, uncertain repair and sparse local validation. Individual ownership of drones or specialised machinery may produce low utilisation and debt; FPO, cooperative, SHG or custom-hiring services can spread fixed cost and provide trained operators. Precision also creates ecological and governance risks. Miscalibration can damage crops, opaque models can misdiagnose local conditions, and lower water cost per hectare may induce crop or area expansion, offsetting aquifer savings. Procurement should measure uptime, calibrated service, per-acre cost and farmer retention rather than assets purchased. Policy should support shared services, open standards, local agronomy, transparent pricing, maintenance networks, farmer feedback and resource governance. Success is net income, resilience and total resource outcome, not equipment distributed.

MAINS 4 - 15 MARKS

Question: Examine how digital technology can improve agricultural credit and crop insurance without creating new exclusion. Answer in 250 words.

Native-body ceiling: 250 words under the required alphanumeric regex.

Model answer:

Digital registries, crop surveys, bank records, weather stations and remote sensing can lower verification cost in agricultural finance. For KCC, reusable borrower and crop information may shorten processing and align limits with seasonal cash flow, while the bank retains underwriting responsibility. Under PMFBY, digital enrolment, CCE recording, YES-TECH and WINDS can improve auditability, yield evidence and claim settlement. These gains are conditional. Farmer ID is not title or loan entitlement. Alternative-data scores may penalise tenants, minor crops or regions absent from training data, and platform-linked credit can encourage over-indebtedness. Insurance models retain basis risk because area yield, weather index or remote estimate may differ from parcel loss. Regulators should monitor rejection, pricing, delinquency, claim reversal and grievance patterns across farm groups. A farmer-centric system needs explicit consent or other lawful basis, purpose limitation, affordability assessment, reasons for adverse decisions, ground verification and human appeal. Offline and assisted routes must cover connectivity and literacy gaps. Technology should supplement prudent banking and transparent insurance, not replace judgment with an unchallengeable score.

MAINS 5 - 20 MARKS

Question: Critically analyse the role of digital platforms and agritech business models in agricultural markets. Answer in 250 words.

Native-body ceiling: 250 words under the required alphanumeric regex.

Model answer:

Digital platforms can reduce search and coordination costs by matching farmers, FPOs, input suppliers, machinery providers, warehouses, transporters, lenders and buyers. e-NAM illustrates the digital-plus-physical model: online discovery and bidding require trusted assaying, aggregation, logistics, settlement and dispute resolution. Agritech firms may use subscriptions, transaction commissions, service fees, input margins or enterprise contracts. Network effects can deepen matching and data-driven services, while shared-service models spread drone or machinery fixed costs across small farms. Yet scale can create new gatekeepers. Opaque ranking, tied credit and inputs, exclusive data access, switching costs and weak bargaining may transfer rents from farmers to platforms. App downloads, registered users or cumulative trade cannot establish price realisation or viable unit economics. Evaluation should compare realised prices, fees, repeat use, service cost, complaints and farmer switching. Policy should preserve open standards, portability, multi-homing, transparent commissions and ranking, competition enforcement and accountable

grievance. FPOs can aggregate demand and supply but need working capital and internal governance. The objective is contestable coordination that improves farmer net returns, not digital intermediation as an end in itself.

MAINS 6 - 20 MARKS

Question: Design an inclusive and accountable digital-agriculture strategy for India. Answer in 250 words.

Native-body ceiling: 250 words under the required alphanumeric regex.

Model answer:

India should adopt a problem-first architecture. Public rails such as AgriStack can provide federated, interoperable identity, map and crop records, while Krishi-DSS supports geospatial planning. Applications should then solve named production, risk, logistics or market failures. Inclusion requires separating recorded owner from actual cultivator, recognising state tenancy arrangements, and offering assisted, offline, voice and local-language channels through field staff, CSCs, FPOs and SHGs. Data governance should specify purpose, minimum collection, access controls, retention, security, correction and audit under the phased DPDP framework and sector rules. Consequential AI, credit and insurance decisions need local validation, confidence disclosure, human review and appeal. Public procurement should avoid vendor lock-in through open standards, portability and exit rights. Shared services should replace indiscriminate gadget distribution where fixed costs are high. States need staff, cadastral quality, seasonal survey capacity and published correction timelines. Evaluation must move from outlay, IDs, surveys and downloads to active adoption, accuracy, net income, risk reduction, resource effects, complaints, reversals and distribution by gender, tenancy, farm size and region. Digital agriculture should expand farmer capability and resilience without automating existing inequalities.

FINAL PRACTICE CHECKLIST

- Did the answer distinguish the technology, data layer, authority and programme stage?
- Did it trace a farmer decision through an implementable action?
- Did it name the weakest physical or institutional complement?
- Did it include exclusion, model, cyber or platform risk and a usable remedy?
- Did it evaluate net income, risk, resources and farmer control rather than app counts?